Sai Sri Harsha Chakravarthula

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Education

Ph.D. in Computer Science and Engineering *Aug. 2024 – Expected May 2027*
University of North Texas Denton, TX
Focus: Cybersecurity in Emerging Systems, AI/ML Security, Secure Edge AI, LLM Security, Cloud and IoT Security, Advanced Algorithms

Master of Science in Artificial Intelligence *Jan. 2022 – Dec. 2023*
University of North Texas Denton, TX
GPA: 3.73/4.00
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Software Engineering, Data Structures, Algorithms, Big Data, Feature Engineering, Data Visualization, and Empirical Analysis

Bachelor of Engineering in Electronics and Communication Engineering *Sep. 2015 – May 2019*
Osmania University Hyderabad, India
Classification: First Class with Distinction

Research Interests

- Artificial Intelligence and Machine Learning; Deep Learning; Large Language Models; Trustworthy and Explainable AI; AI/ML Security; Secure Edge AI; Internet of Medical Things; Healthcare AI; Computer Vision; Anomaly Detection; Cloud and IoT Security; Privacy-Preserving AI; Hardware-Rooted Trust; Intelligent Decision-Support Systems.

Professional and Research Experience

Doctoral Researcher *Aug. 2024 – Present*
University of North Texas Denton, TX

- Conduct research in artificial intelligence, machine learning, trustworthy LLM systems, secure edge computing, healthcare AI, and intelligent IoT/IoMT applications.
- Developed AI and security frameworks involving LLM-output verification, anomaly detection, edge intelligence, cryptographic integrity mechanisms, blockchain-based provenance, and hardware-rooted trust.
- Designed and evaluated machine-learning and deep-learning models using Python, PyTorch, TensorFlow, Scikit-learn, NumPy, and Pandas.
- Conducted data preprocessing, feature engineering, statistical analysis, model training, validation, and performance evaluation.
- Investigated AI-driven healthcare applications including food–medication interaction screening, intelligent risk prediction, IoMT systems, and anomaly detection.
- Contributed to peer-reviewed journal and conference research in artificial intelligence, machine learning, LLM security, edge intelligence, and healthcare AI.

Software Engineer *Feb. 2024 – Aug. 2024*
TEKSOLVE IT SOLUTIONS LLC Remote, United States

- Developed and maintained backend and full-stack software components using Java, JavaScript, Node.js, Spring Boot, REST APIs, React, and database-driven systems.
- Collaborated on software design, debugging, testing, feature implementation, and application maintenance.

HSCC Peer Mentor / Data Assistant - Student Outreach Program *2022 - 2024*
University of North Texas Denton, TX

- Served in a university outreach program supporting middle-school and high-school students through academic and career-focused mentoring.
- Visited local middle schools and high schools to explain academic pathways, course options, college preparation, and career-development opportunities.
- Mentored university students and provided academic, technical, and professional support.
- Assisted students with academic planning, university resources, technical questions, and strategies for academic success.
- Guided students in understanding how course choices, technical skills, and academic interests could support future college and career goals.
- Supported mobile and web application development using Java, JavaScript, XML, Android Studio, and related technologies.
- Assisted with student-data analysis and structured administrative workflows using Excel and Macros.

- Communicated technical and academic information to students from different educational and technical backgrounds.

Programmer Analyst *Jan. 2020 - Dec. 2021*
Cognizant Technology Solutions Chennai, India

- Developed enterprise and backend software using Java, JavaScript, SQL, REST APIs, Oracle, AWS, and Microsoft Azure.
- Analyzed banking datasets and extracted features for predictive and risk-analysis applications.
- Worked with relational and NoSQL databases, including Oracle, MongoDB, Cassandra, and DynamoDB.
- Developed PL/SQL procedures, packages, triggers, and views for enterprise systems.
- Supported automated builds and deployments using Git, Jenkins, Terraform, and CI/CD workflows.
- Worked with Azure Web Apps, Azure Data Factory, Azure Data Lake, Grafana, and Prometheus.

Systems Engineer *Jan. 2019 - Dec. 2019*
Raster Technologies Hyderabad, India

- Developed applications and automation scripts using Python, SQL, PL/SQL, AutoLISP, HTML, CSS, and related technologies.
- Worked with IRX and AutoLISP for engineering, automation, and technical application workflows.
- Developed database procedures and automated data-processing and reporting workflows.
- Automated Excel-based processes using Macros and VBScript.

Teaching Experience

Graduate Teaching Assistant *Fall 2024 - Present*
University of North Texas Denton, TX

CSCE 1010 - Introduction to Computer Science
Spring 2026: Sections 501, 551; Fall 2026: Sections 501, 551, 552

- Supported introductory computing instruction involving computational thinking, programming, algorithms, Python, problem solving, databases, networking, and foundational computing concepts.
- Assisted students with laboratories, programming assignments, debugging, algorithm development, and implementation.
- Graded labs, quizzes, assignments, and course activities using established rubrics and provided constructive feedback.
- Conducted office hours and provided individualized support to students with different levels of programming experience.

CSCE 4010 - Social Issues in Computing
Section 401, Summer 2026

- Supported coursework addressing ethical, professional, social, and societal issues related to computing technologies.
- Graded coursework using established rubrics and conducted scheduled office hours for student support.

CSCE 4901 - Software Development Capstone I
Section 502, Spring 2026

- Supported senior software-development teams in project planning, requirements analysis, implementation, documentation, teamwork, and milestone tracking.
- Evaluated individual and team deliverables and provided project guidance through student meetings and office hours.

CSCE 3055 - IT Project Management
Section 501, Spring 2026

- Supported coursework involving project planning, software-project management, documentation, teamwork, and project-delivery processes.
- Assisted with grading, assignment management, course administration, and student questions.

CSCE 2100 - Foundations of Computing
Sections 003 and 004, Fall 2025

- Supported students with computing foundations, algorithmic problem solving, programming concepts, and technical coursework.
- Assisted with grading, office hours, assignment support, and clarification of course concepts.

CSCE 5150 - Analysis of Computer Algorithms
Section 002, Fall 2024

- Supported graduate-level coursework involving algorithm design, analysis, complexity, and computational problem solving.
- Assisted with assignments, grading, course activities, and student questions involving algorithmic concepts.

Publications

Published

- **S. S. H. Chakravarthula**, I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iMediFood-Shield: Secure Edge AI for Food and Medication Interaction Screening," *Electronics*, vol. 15, 2026. DOI: 10.3390/electronics15173799.

- L. Rachakonda, **S. S. H. Chakravarthula**, S. P. Mohanty, and E. Kougianos, "cStick 2.0: An IoMT-Edge Based Vision-Enabled Smart System for Personalized Fall Prediction and Detection," *Electronics*, vol. 15, 2026. DOI: 10.3390/electronics15153375.
- **S. S. H. Chakravarthula**, A. Mitra, S. P. Mohanty, and E. Kougianos, "Deepfake Audio Detection: Differentiating Modulated and Deepfake Audio," *IEEE MetroCon*, 2025.
- M. V. Naresh, A. Akshith, **C. H. Sai Sri Harsha**, and M. Sai Kumar, "Visiting Center Based Portable Emissary Approach for Data Gathering in Wireless Sensor Networks," *2019 3rd International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 1023-1028, 2019. DOI: 10.1109/ICCMC.2019.8819746.

Accepted

- **S. S. H. Chakravarthula**, V. K. V. V. Bathalapalli, V. P. Yanambaka, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "LLMTrustChain: A Hardware-Rooted Blockchain Framework for Tamper-Evident Verification of LLM Outputs," *IFIP IoT 2026*.
- **S. S. H. Chakravarthula**, V. P. Yanambaka, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "EdgeShield-LLM: Hardware-Rooted Integrity Verification for Edge LLM Deployment Bundles," *ISVLSI 2026*.
- **S. S. H. Chakravarthula**, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "SANDS: Security-Aware Anomaly Detection for IoMT Systems," *ISVLSI 2026*.
- I. D. Siripurapu, **S. S. H. Chakravarthula**, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iMediFood-AI: Risk-Aware Dietary–Medication Interaction Prediction," *OCIT 2026*.
- **S. S. H. Chakravarthula**, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "SideShield: Runtime Side-Channel Leakage Detection for Edge LLMs," *iSES 2026*.

Professional Service & Peer Review

Peer Reviewer

- *SN Computer Science*, Springer Nature, 2026.
- IEEE International Symposium on Smart Electronic Systems (iSES), 2026.
- International Conference on Emerging Technologies and Computing (OCIT), 2026.
- IEEE Computer Society Annual Symposium on VLSI (ISVLSI), 2026.
- IFIP Internet of Things Conference (IFIP IoT), 2026.

Selected Research Projects

LLMTrustChain - Trustworthy LLM Systems

- Developed a hardware-rooted framework for tamper-evident verification of LLM-generated outputs using cryptographic hashing, edge AI, and blockchain provenance.

iMediFood-Shield - Healthcare AI

- Developed a secure edge-AI framework for food–medication interaction screening using machine learning, risk classification, and model evaluation.

SANDS - Security-Aware Anomaly Detection

- Developed machine-learning-based anomaly detection approaches for security monitoring in IoMT systems.

Deepfake Audio Detection

- Applied machine learning, audio feature extraction, preprocessing, and classification techniques to distinguish manipulated and deepfake audio.

Bank Churn Prediction Using Neural Networks

- Developed and evaluated machine-learning and artificial-neural-network models using Python, NumPy, Pandas, Scikit-learn, and TensorFlow, achieving approximately 95% classification accuracy.

Facial Recognition System

- Developed a real-time facial-recognition application using Python, Django, TensorFlow, GCP, and computer vision technologies.

Current and Ongoing Research

- Develop **BlockShield** for strengthening integrity, authentication, and trust in edge and emerging computing systems using blockchain and hardware-rooted security mechanisms.
- Investigate **Physical Unclonable Function (PUF)-based security** for device authentication, hardware identity, key generation, and tamper-resistant protection of edge and IoT devices.
- Advance **LLM security for edge environments** through integrity verification, secure deployment, provenance tracking, tamper detection, and protection of models and generated outputs.
- Explore **RAG-based edge security** for detection and mitigation of security threats in distributed and resource-constrained edge

systems.

Technical Skills

Programming: Python, Java, C, C++, JavaScript, SQL, PL/SQL, MATLAB, AutoLISP, BASH

Deep Learning & AI: Machine Learning, Deep Learning, Neural Networks, CNNs, LSTMs, Transformers, Large Language Models (LLMs), Computer Vision, Natural Language Processing, Explainable AI, Federated Learning

Tools & ML Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, PySpark, NLTK, Matplotlib, Jupyter, Google Colab, PyCharm, Visual Studio Code, Git, IRX

Software Development: Django, Spring Boot, Node.js, React, REST APIs, Agile/Scrum, Debugging, Testing, Technical Documentation

Databases & Cloud: Oracle, MySQL, SQL Server, MongoDB, Cassandra, DynamoDB, AWS, Microsoft Azure, Google Cloud Platform (GCP)

DevOps & Systems: Git, Jenkins, Terraform, CI/CD, Linux, Edge Computing, IoT/IoMT

Industrial Automation: CNC Machine Operation, CNC Programming, PLC Programming, PLC Control Logic, Industrial Automation

Internships

Engineering Intern / CNC & PLC Technical Trainee *Undergraduate Internship*
Bharat Heavy Electricals Limited (BHEL) Hyderabad, India

- Completed hands-on technical training involving Computer Numerical Control (CNC) machines and Programmable Logic Controllers (PLCs).
- Worked with CNC machine programming and operation, gaining practical exposure to automated manufacturing and machine-control processes.
- Gained hands-on experience with PLC programming, control logic, and operation of PLC-based industrial systems.
- Developed practical understanding of how programmed control systems are used to operate and automate industrial machinery.

Awards and Honors

- **Mission Critical Award**, Cognizant Technology Solutions.
- **Best Employee Award**, Cognizant Technology Solutions.
- **Employee of the Month**, HSCC Peer Mentor / Data Assistant, University of North Texas.

Professional Development and Memberships

- **UNT SEED Program Participant**, University of North Texas, Fall 2026. Professional development in college teaching, course design, measurable learning outcomes, active learning, assessment, and evidence-based instructional practices.
- **Certified International Teaching Assistant (ITA)**, University of North Texas, Spring 2026. Completed the UNT International TA Training Program and communication workshops for instructional-support responsibilities.
- **IEEE Student Member**
- **IEEE Young Professionals Member**

Leadership and Service

- **Team Leadership, Cognizant Technology Solutions**
  Served in a team-lead capacity for project activities, coordinating development tasks, supporting team members, resolving technical issues, and facilitating communication across project stakeholders.
- **Team Leadership, HSCC Peer Mentor / Data Assistant Program, University of North Texas**
  Helped coordinate peer-mentoring and outreach activities, guided team tasks, supported student engagement, and contributed to academic and career-readiness outreach for middle-school and high-school students.
- **Student Mentoring and Outreach, University of North Texas**
  Mentored students on academic pathways, course selection, college preparation, technical skills, and career-development opportunities.

Teaching Interests

- Introduction to Computer Science; Programming Fundamentals; Python; C/C++; Java; Data Structures; Algorithms; Artificial Intelligence; Machine Learning; Deep Learning; Software Engineering; Database Systems; IT Project Management; Cloud Computing; Cybersecurity; Social and Ethical Issues in Computing; Software Development Capstone; Project-Based Computing Courses.